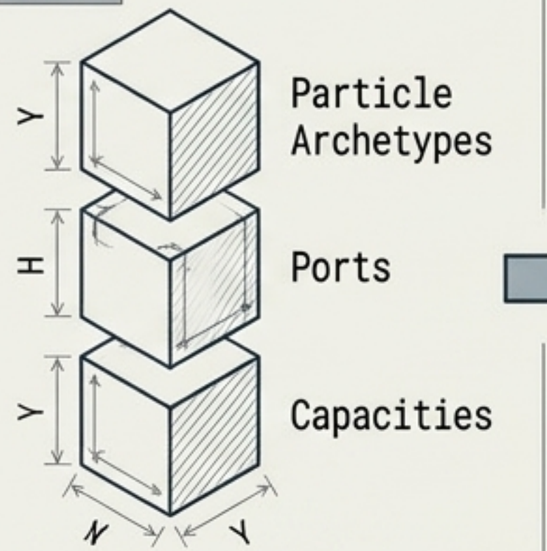
WHAT IS IGSPA?

Information-Geometric Stochastic Particle Assembly (IGSPA) is a framework for simulating interactive particle systems with no underlying spatial geometry.

It achieves this by combining:

1. Gillespie SSA with global-bucket optimization.
2. Fisher Information Geometry.
3. Explicit multi-graph topology tracking.

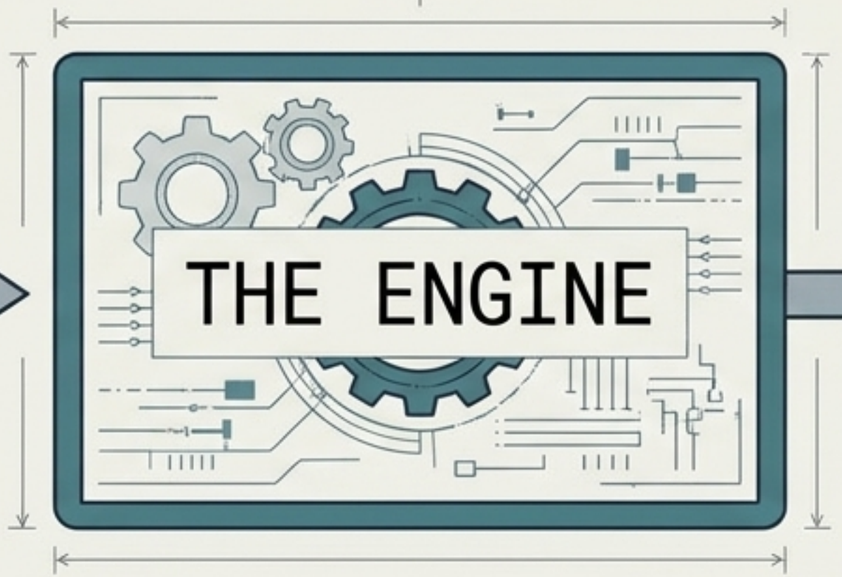
INPUTS



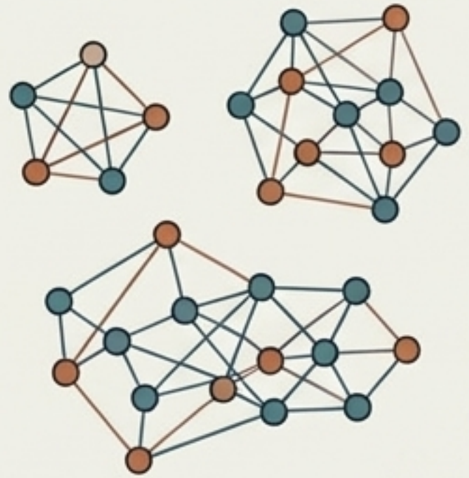
Particle Archetypes

Ports

Capacities



OUTPUTS



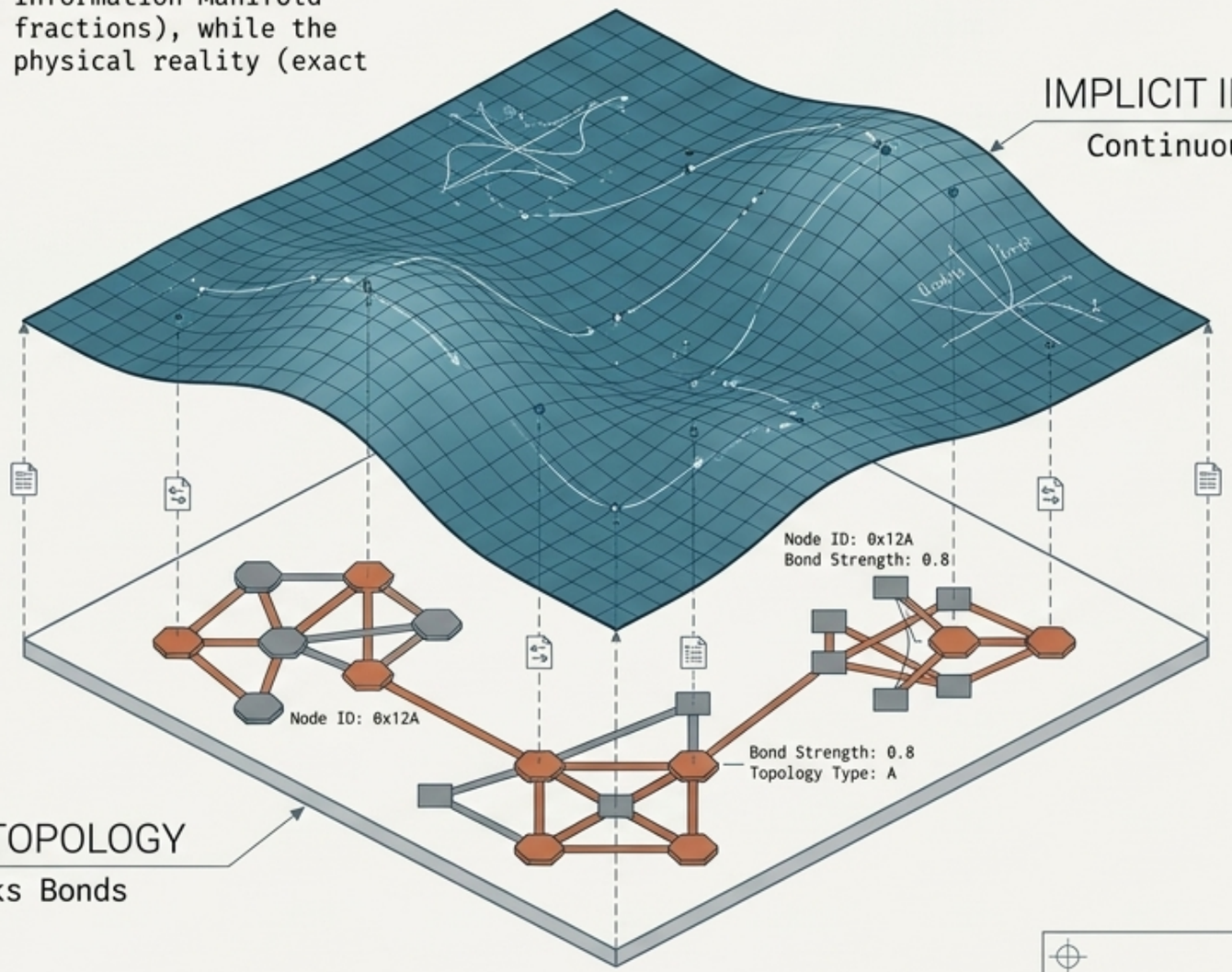
Complexes & Geometries

PROJECT:	DATE:	DRAWING NO:	REV:
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THE DUAL REPRESENTATION ARCHITECTURE

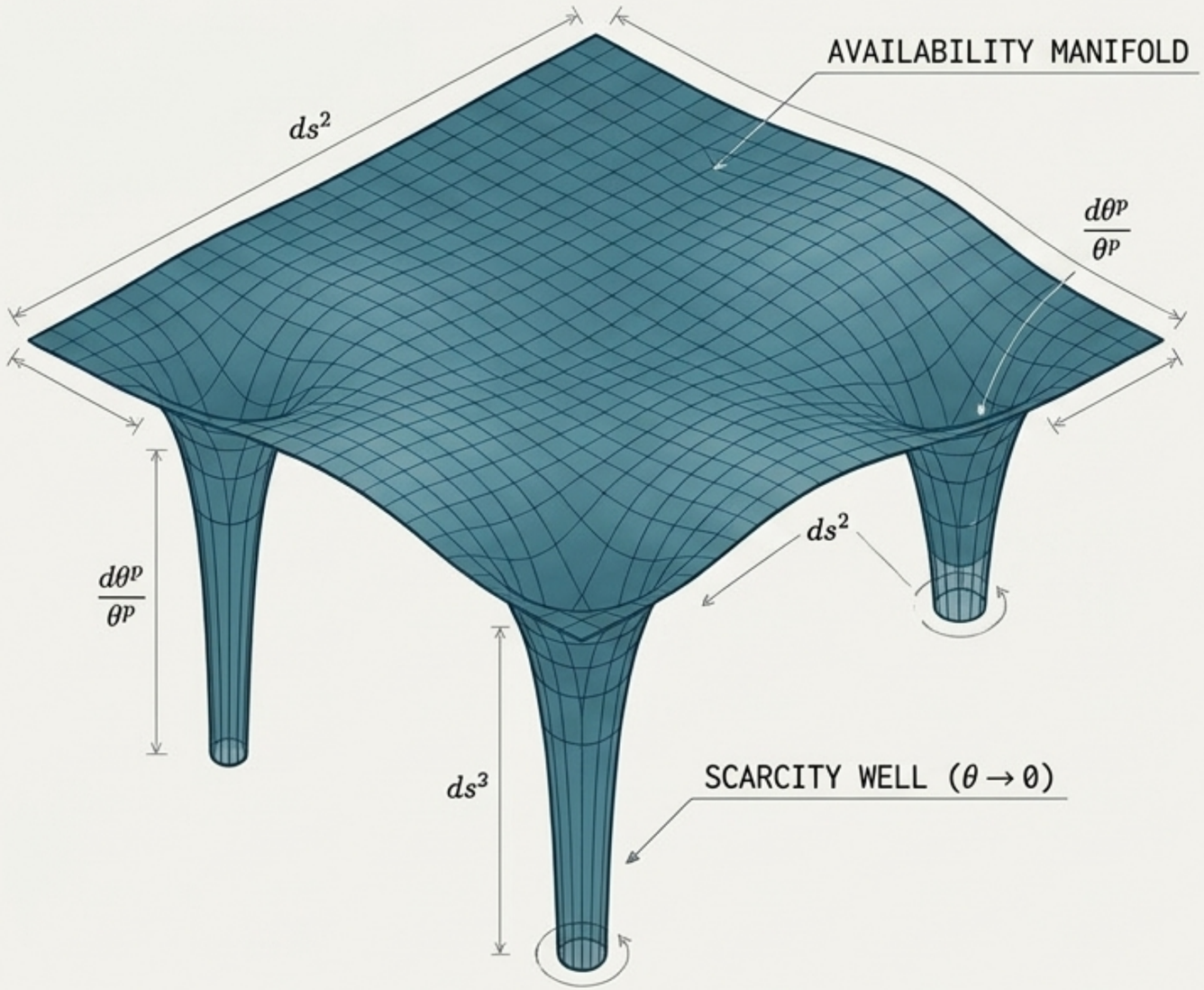
IGSPA abandons Cartesian space for a synchronized dual representation. The continuous Information Manifold tracks potential (availability fractions), while the discrete Explicit Graph tracks physical reality (exact bond structure).



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REPRESENTATION 1: THE IMPLICIT INFORMATION MANIFOLD



COORDINATE TRACKING

Tracks availability fractions $\theta_{ip} \in [0, 1]$ for each particle and port.



FISHER INFORMATION METRIC

Distance is governed by the Shahshahani form:

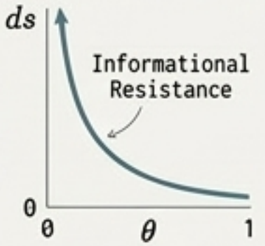
$$ds^2 = \sum_{\text{Information Distance}} \frac{(d\theta^p)^2}{\theta^p}$$

Availability Fraction

Availability Fraction

SCARCITY GEOMETRY

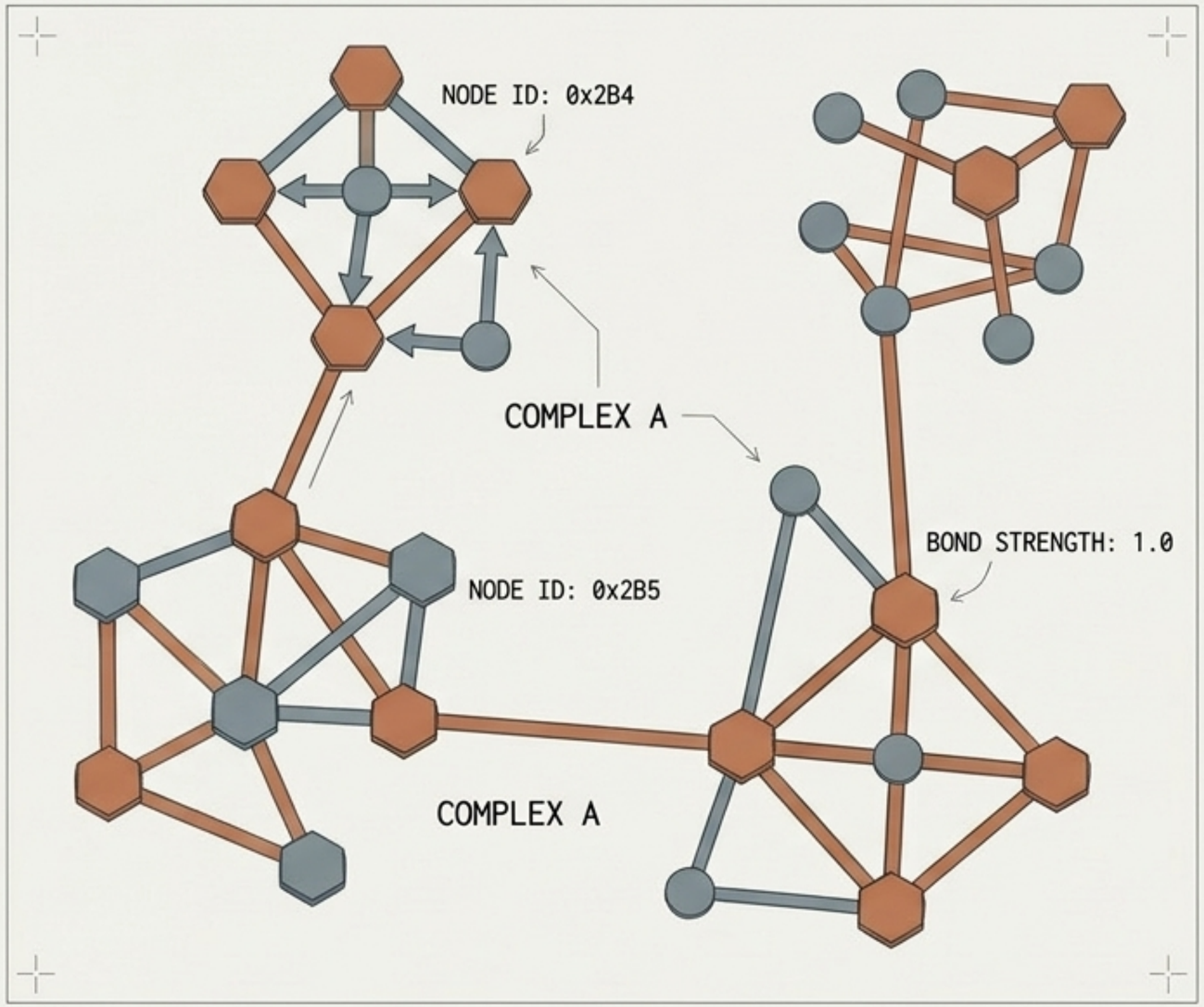
As a port resource approaches exhaustion ($\theta \to 0$), the informational distance stretches to infinity. This geometrically encodes extreme thermodynamic resistance, perfectly mirroring the physical difficulty of rare-state updates.



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REPRESENTATION 2: THE EXPLICIT GRAPH TOPOLOGY



STRUCTURE DEFINITION

A combinatorial multi-graph G(t) = (V, E(t)) tracking exact bond structures and physical complexes via connected components.

THE CONSERVATION LAW

$$\theta_{ip}(t) = 1 - \frac{\sum_{j \in V} A_{ijp}}{K_{ip}}$$

The explicit graph strictly bounds the implicit manifold.

EXACT MASS CONSERVATION

The explicit graph ensures exact mass conservation and allows for precise complex detection without ever calculating spatial proximity.

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	DRAWING NO: 06-EXPLICIT_GRAPH	REV: A

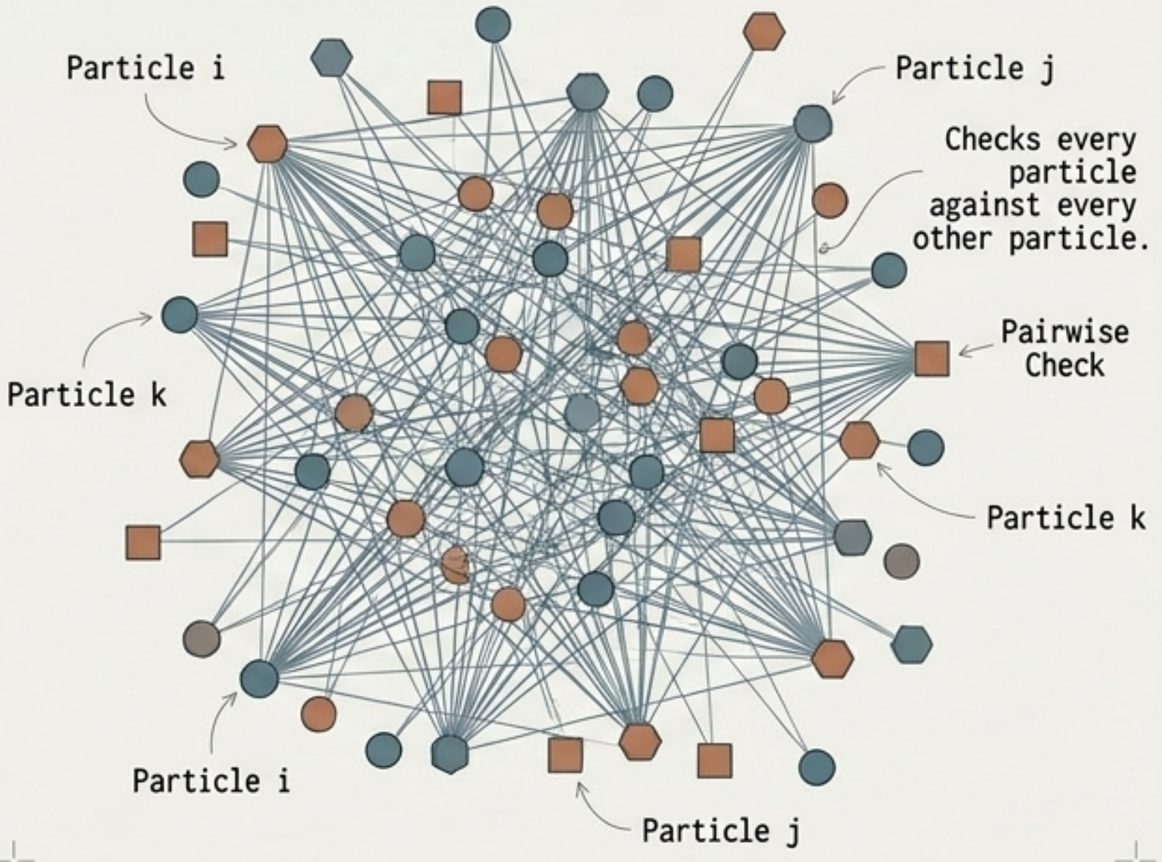
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THE ENGINE'S SECRET SAUCE: GLOBAL BUCKET OPTIMIZATION

Instead of looping over $O(N^2)$ pairs, IGSPA maps active particles into global containers: $B_p = \{i \mid \theta_{ip} > 0\}$. By evaluating the whole system's binding potential at once, pairwise comparisons are completely eliminated.

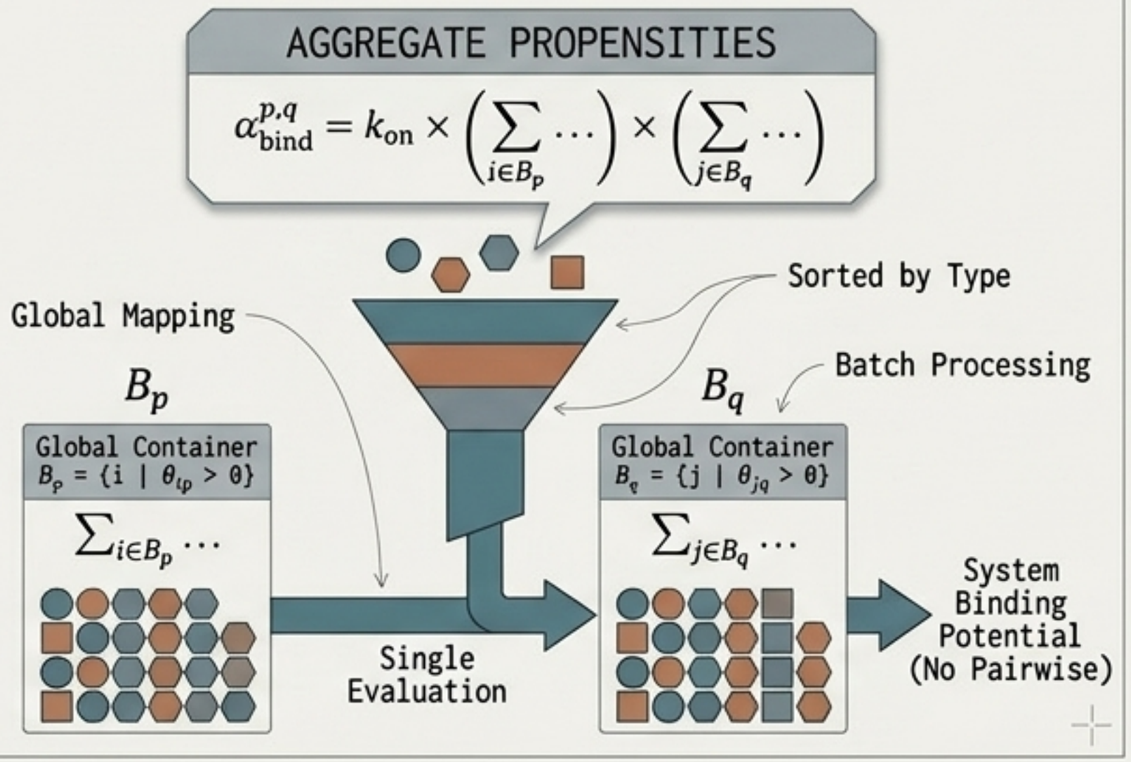
STANDARD SSA

Quadratic Overhead



IGSPA

Global Buckets



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	DRAWING NO: 07-GLOBAL_BUCKETS_OPT	REV: A

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IGSPA vs. Standard SSA: The Performance Shift

METRIC	STANDARD SSA	IGSPA ENGINE
Time / Step	$O(N^2\|P\|)$	$O(\|P\| + \|E\|)$
Space	$O(N^2)$	$O(N)$
Scaling	Quadratic	Linear / Constant

A fundamental shift from quadratic bottlenecks to linear efficiency, enabling the scale required for closed-loop machine learning.

PROJECT: COMPLEXITY SIMULATION

DATE: 2024 | DRAWING NO: 08-PERFORMANCE_SHIFT | REV: A

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